

ENHANCING SINGLE-CELL RNA-SEQ ANALYSIS THROUGH THE INTEGRATION OF UNACCOUNTED INTERGENIC REGIONS

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1. LIST OF ABBREVIATIONS

cDNA	complementary DNA
CPM	counts per million
GRN	gene regulatory network
NGS	next generation sequencing
RT	reverse transcription
scRNAseq	single cell RNA sequencing
UMI	unique molecular identifier

2. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) is becoming an increasingly popular tool for analyzing cellular systems. This technology enables whole transcriptome sequencing of thousands of cells in a single experiment, generating a vast amount of data. The typical workflow for analyzing such data involves mapping reads to a known genome reference and constructing cell-gene matrices, i.e. matrices in which rows correspond to cells, columns correspond to genes, and each entry indicates the expression level of a gene in a specific cell. Such matrices are used in downstream analyses. However, some reads are always mapped to the genome but remain unassigned to any known gene. Hence, such reads are not used in downstream analysis.

Since these unassigned reads can constitute a significant fraction of the total reads (often up to 30%), understanding the reasons behind their origin could help improve scRNA-seq technologies and data analysis.

There are two potential sources of unassigned reads: either they are sequencing artifacts, or they originate from real transcripts that remain unassigned due to issues related to transcriptomic references.

The two main challenges with transcriptomic references are:

1. Incomplete transcriptome annotations – even though there are given great efforts to annotate all genes, it is very likely that not all genes are annotated, and many remains to be found. Such yet undefined genes are missed in the typical scRNAseq analysis.
2. Complexity and overlaps in transcriptomic features – the human (and many other species) genome is very complex, with many overlapping features. In such cases, mapping algorithms assign multiple features for a given read, and such reads are usually discarded from the downstream analysis.

This project focuses on unassigned reads in scRNA-seq datasets, aiming to uncover their origins and enhance scRNA-seq data analysis by incorporating biologically meaningful data that is typically disregarded.

3. AIM AND TASKS

Aim The aim of this project is to investigate whether it is possible to improve scRNA-seq analysis by including unassigned reads in the downstream analysis.

Tasks

1. Identify genomic regions containing unassigned reads, and classify them as either intersecting with genes or intergenic.
2. For intergenic regions, collect evidence that would confirm them being noise or unannotated genes.
3. For intersecting regions, attempt to resolve problems related to the transcriptomic reference which lead to reads being unassigned.

4. LITERATURE REVIEW

4.1 Introduction to single cell transcriptomics

Cells are the fundamental units of life, forming the basis of all living organisms. One of the major goals of biology is to understand cellular systems and the processes occurring within cells. Since the discovery of the DNA structure in 1953 and the development of the conceptual framework for genetic information transfer (Watson and Crick 1953), scientists have made significant efforts to sequence the genomes of various organisms. This led to the development of the first sequencing methods, such as Sanger sequencing in 1975 (Sanger, Nicklen, and Coulson 1977), which laid the foundation for next-generation sequencing (NGS) technologies in use today, such as the widely used Illumina platform (Heather and Chain 2016). Current sequencing methods allow us to obtain the complete genetic sequence of any organism. However, the genome alone cannot explain the full diversity of cells in multicellular organisms, as all cells share the same genome but exhibit significant variation in shape, size, and function.

RNA sequencing (RNAseq), on the other hand, enables the measurement of gene expression within cells, providing valuable insights into cellular processes. RNAseq methods largely follow DNA sequencing protocols, with the addition of a step where complementary DNA (cDNA) is synthesized from RNA (Heumos et al. 2023). The first RNAseq methods were developed for bulk sequencing, where RNA from entire cell populations is sequenced, providing an average gene expression profile across the population. Although bulk RNAseq has provided valuable insights into the dynamics of cellular processes (such as changes in disease states in response to therapeutics, detection of gene isoforms, gene fusions, and various other properties of target cells (Heumos et al. 2023)), this approach masks non-dominant processes and cell-to-cell variability through averaging. This limitation was addressed by the introduction of single-cell RNA sequencing (scRNAseq) methods, which allow for the generation of transcriptomic profiles from individual cells, providing high-resolution insights into cellular systems.

Current scRNAseq methods enable the generation of transcriptomic profiles from thousands of cells at unprecedented resolution in a single experiment. These data can be used for constructing cellular atlases (Rozenblatt-Rosen et al. 2017), understanding disease mechanisms (Z. Zhang, Chen, and X. Peng 2024), exploring cell differentiation and developmental processes (Skinner, Asad, and Haque 2024), among many other applications.

4.2 scRNAseq data generation and analysis

4.2.1 Overview of scRNAseq protocols

The generation of scRNA-seq data is a complex, multi-step process that varies across different protocols. These protocols can be grouped based on several criteria, such as the type of RNA capture (e.g., 3' end, 5' end, or full-length) or the method of cell isolation (e.g., droplet-based methods such as inDrops (Klein et al. 2015), Drop-seq (Macosko et al. 2015), and Chromium by 10X Genomics (Zheng et al. 2017), or plate-based methods such as CEL-Seq2 (Hashimshony et al. 2016) and Smart-seq2 (Picelli et al. 2013)).

In this project, datasets generated using droplet-based 3' end sequencing methods were analyzed. Therefore, these methods will be the focus of the following literature review.

3' End Sequencing and Polyadenylation 3' end sequencing captures RNA molecules using primers complementary to poly(A) tails. Polyadenylation at the 3' end is a post-transcriptional modification in which non-templated adenosines are added to the 3' end of mRNA molecules. Although the poly(A) tail is present in almost all mRNAs, its length varies and can influence mRNA fate, stability, and translation efficiency (Brouze et al. 2022).

3' end sequencing protocols exploit this feature to selectively capture RNA molecules, enabling high-throughput and cost-effective sequencing. However, some RNA molecules lack poly(A) tails and are therefore not captured by these methods, such as replication-dependent histone mRNAs (Brouze et al. 2022).

Common Steps in scRNA-seq Protocols Despite differences between scRNA-seq protocols, they share key steps: single-cell isolation, library preparation, and sequencing (Andrews and Hemberg 2018). In droplet-based methods, single cells are encapsulated in individual droplets containing hydrogel-attached barcoding primers and a lysis mix (an example of a droplet generation device is shown in Figure 4.1). Primers used in these protocols typically share a common structure, including:

- **Cell barcodes:** Unique sequences that identify the cell from which a particular read originates (since all droplets are pooled and sequenced together).
- **Unique molecular identifiers (UMIs):** Short sequences used to quantify the original number of RNA molecules, helping to eliminate amplification bias.
- **PCR handles:** Sequences that facilitate amplification.
- **Poly-T sequences:** In 3' end sequencing methods are used to selectively capture polyadenylated RNAs (X. Zhang et al. 2019).

An example of primer design is shown in Figure 4.2. Once cells are encapsulated in droplets, lysis occurs, releasing RNA, which is then captured by the primers.

Depending on the method, reverse transcription may occur within the droplets (as in inDrops and 10X Genomics) or in bulk (as in Drop-seq). Subsequent steps typically include RNA fragmentation, PCR amplification, and next-generation sequencing (NGS).

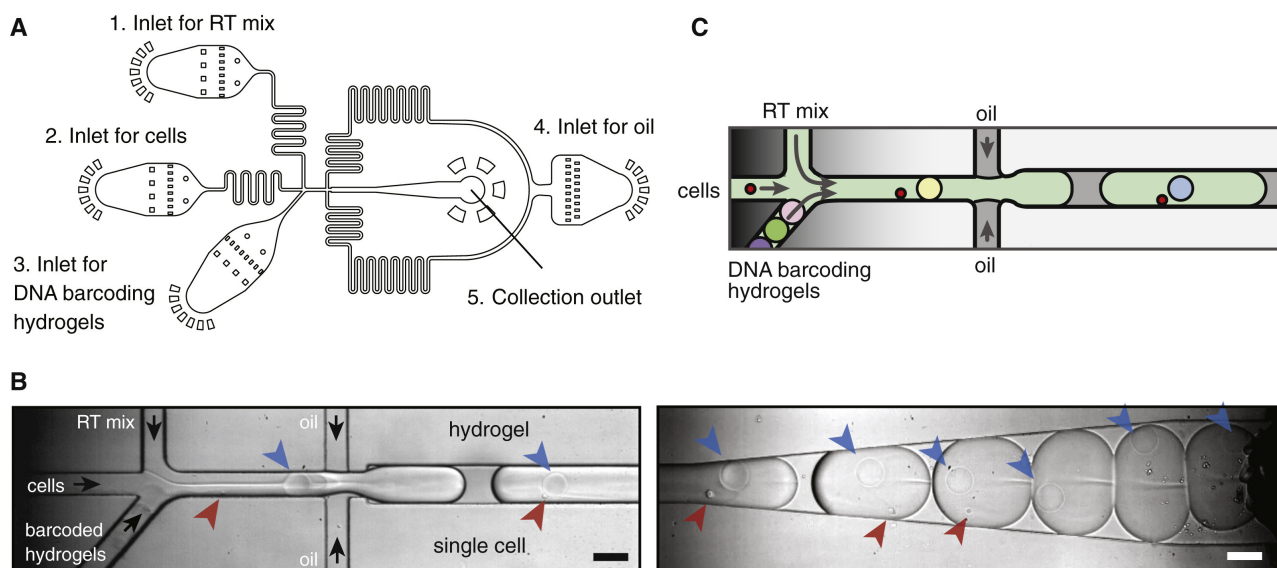


Figure 4.1: An example of microfluidics device for droplet generation (inDrops). A) Schematic view of the device. B) Snapshots of droplet generation and collection. C) Scheme of droplet generation. Figure taken from Klein et al. (2015).

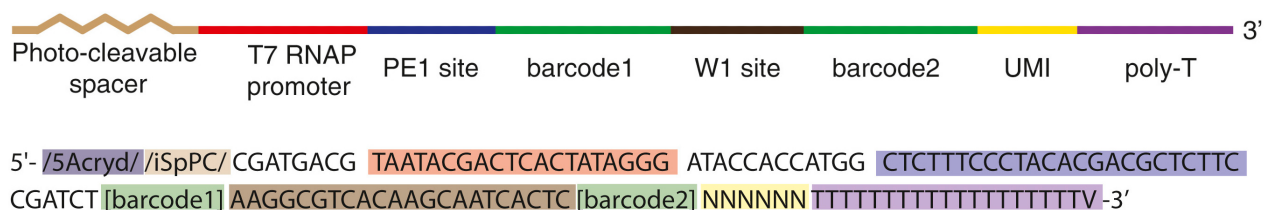


Figure 4.2: Example of primer design (inDrops). The image above shows schematic view, while below is given example with sequences. The structure of primers varies between the protocols, the main parts (UMI site, barcodes, poly-T's) are found in most of them. Here also present are promoter region for RNA polymerase (red), sequencing primer (blue), synthesis adaptor (dark brown). Figure taken from Klein et al. (2015).

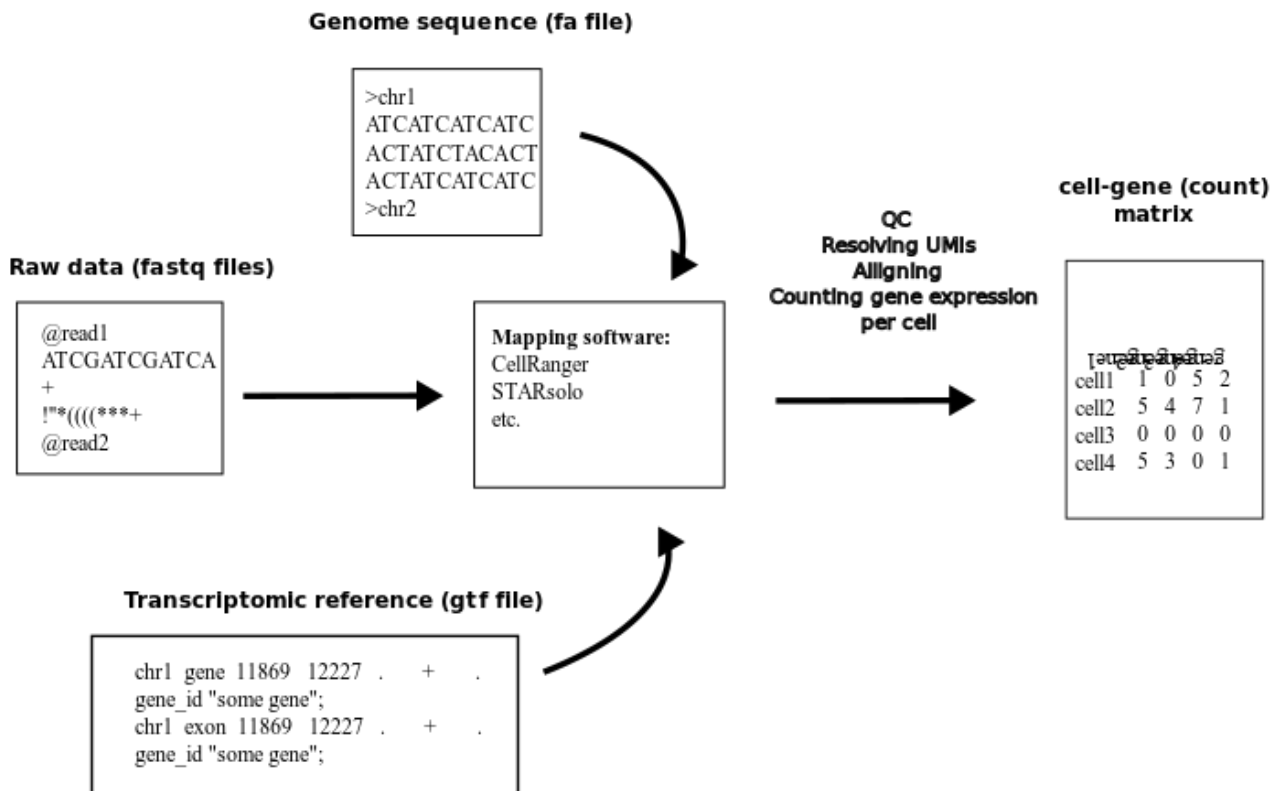


Figure 4.3: Inputs and outputs of mapping software.

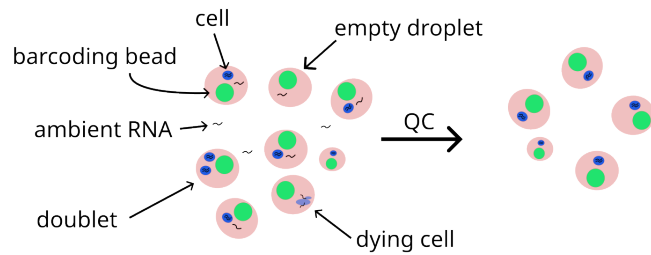
4.2.2 scRNAseq data analysis

Raw data processing The output from NGS machines is typically converted to FASTQ files, containing recorded sequences, as well as (depending on method) barcode and UMI sequences, and quality scores. The subsequent processing steps include quality control of FASTQ file (based on quality scores), filtering duplicate reads (using UMIs), mapping reads to the genome sequence, assigning the reads to the genes, and finally, counting gene expression per cell (barcode) (Heumos et al. 2023) (see figure 4.3). Usually, all these steps are performed with a single piece of dedicated software, such as STARsolo (Kaminow, Yunusov, and Dobin 2021), CellRanger (Zheng et al. 2017) or other. It should be noted, that there are variations in the pipeline described above, depending on many experiment-related (e.g., whether the genome sequence or transcriptome of the study organism is known), or method-related (e.g., whether UMIs are used in the protocol) factors. The typical result of such processing is cell-gene matrix (i.e., a matrix where rows represent cells, columns represent genes, and each entry indicates the number of captured RNAs for a given gene in a specific cell).

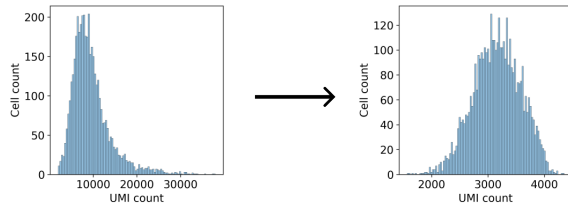
Cell-gene matrix processing The next steps in the analysis of the scRNAseq data involves following steps (see Figure 4.4 for graphical summary):

- **Quality control** The quality of individual cells (represented as all reads harboring a unique barcode) can be evaluated based on several factors, such as mitochondrial gene content (apoptotic cells tend to have a higher proportion of mitochondrial transcripts (Heumos et al. 2023)) or total number of captured genes (very low numbers can be produced by ambient RNA entering empty

Quality control



Normalization



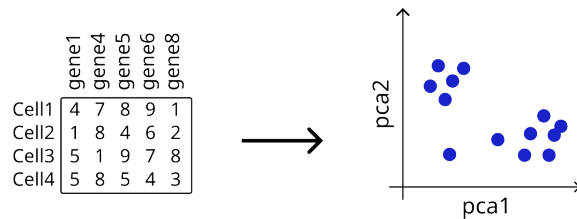
Feature selection



	gene1	gene2	gene3	gene4	gene5	gene6	gene7	gene8
Cell1	4	5	0	7	8	9	0	1
Cell2	1	5	0	8	4	6	0	2
Cell3	5	5	1	1	9	7	0	8
Cell4	5	5	0	8	5	4	0	3

	gene1	gene4	gene5	gene6	gene8
Cell1	4	7	8	9	1
Cell2	1	8	4	6	2
Cell3	5	1	9	7	8
Cell4	5	8	5	4	3

Dimensionality reduction



Clustering and other analyses

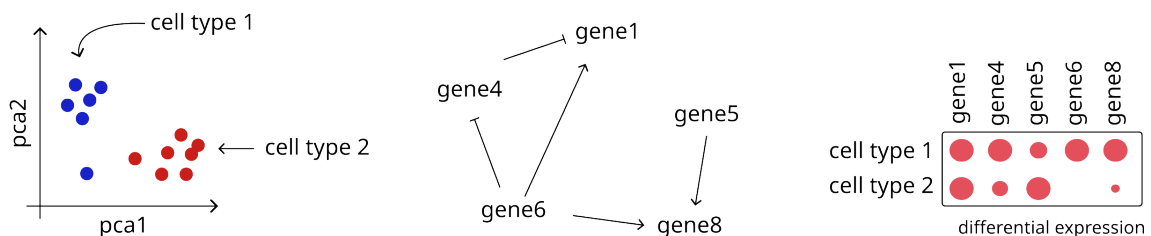


Figure 4.4: The typical data analysis pipeline. Firstly data is cleaned and normalized, then (to reduce dimensionality) non-informative genes are filtered out and dimensionality reduction methods are applied. Final steps depends on the focus of study, but often include cell clustering, annotation, differential gene expression analysis or gene regulatory network inference.

droplets). In some cases, two cells can end up in one droplet, resulting in count matrix row corresponding to genes from both cells. Such matrix entries (doublets) can be filtered by using specialized software such as Scrublet (Wolock, Lopez, and Klein 2019) or scDblFinder (Germain et al. 2022). Another source of noise in scRNAseq data is ambient RNA, which consists of RNA that escapes individual droplets and spreads into the medium or other droplets, leading to background noise. Even though the amount of such RNA is not high (in good quality datasets it can be around 2% (Young and Behjati 2020)), removing these RNAs from the count matrix can improve data quality. This can be achieved by identifying the background noise profile from empty droplets and adjusting the count matrix accordingly. There are dedicated softwares, such as SoupX (Young and Behjati 2020), decontX (Yang et al. 2020), CellBender (Fleming et al. 2023) and others.

- **Normalization** The next step in preprocessing pipeline is normalization. The goal of normalization is to transform the data so that the variation in gene expression levels is comparable, making subsequent analysis more efficient (Ahlmann-Eltze and Huber 2023). Normalization can also help eliminate biases, such as differences in sequencing depth when combining data from multiple samples (Lingen, Suarez-Diez, and Saccenti 2024). There are numerous normalization methods, based on different approaches (e.g., delta-method-based, residual-based, latent gene expression-based, count-based (Ahlmann-Eltze and Huber 2023)). Thus, selecting a normalization method should be done carefully, depending on the experimental design. General recommendations for normalization suggest comparing several methods, and if the results are similar, opting for the simpler method (Lingen, Suarez-Diez, and Saccenti 2024). Sophisticated methods do not necessarily show better results, and a recent benchmarking study by Ahlmann-Eltze and Huber (2023) has shown that simpler method (particularly the logarithm normalization, where each element y of count matrix is transformed by formula $y_{transformed} = \log(y + 1)$) performs as well or better than more advanced methods.
- **Filtering genes** Once the data is normalized and cleaned, one can filter out non-informative genes. Initially, count matrices contain all the genes that are present in the transcriptome. However, not all of them are expressed in the sequenced data, or are expressed in negligible numbers (Heumos et al. 2023). Therefore, it is common practice to filter such genes (e.g., genes that are expressed in less than three cells). Moreover, some genes might be expressed in all the cells more or less evenly (housekeeping genes), which do not provide useful information that could be useful in, for instance, grouping cells or determining cell types. Therefore, in many applications, it is beneficial to leave only those genes, that are highly variable between cells. In such way, the dimensionality of the count matrix is greatly reduced without losing significant information. Additionally, genes that are outside the scope of the specific study can also be filtered out.
- **Dimensionality reduction** Even after filtering and selecting only highly variable genes, several thousand genes usually remain. It is not feasible to visualize (and hard to interpret in general) data of such high dimensionality, therefore, dimensionality reduction is essential step of subsequent analysis. The idea of dimensionality reduction is simple: to reduce the dimensions of the data losing as little information as possible. There are a number of dimensionality

reduction methods based on different mathematical concepts, but the most widely used today include t-SNE (Hinton and Roweis 2002), UMAP (McInnes, Healy, and Melville 2018) and principal component analysis (PCA). Although the use of these algorithms are supported by some benchmarking studies (in the study of Xiang et al. (2021), t-SNE was showed best performance, while UMAP showed the highest stability), other benchmarking studies report different findings. The study of Koch et al. (2021) suggested that such overlooked methods as latent Dirichlet allocation (LDA) and PHATE are among the best performing (metrics for comparison included clustering accuracy, structure preservation, efficiency and others). Meanwhile Sun et al. (2019) provided guidelines for choosing dimensionality reduction method depending on downstream analysis tasks, and in their results UMAP and tSNE were not on the top choices. Thus, while UMAP remains the gold standard method in the field (usually preceded by PCA for initial dimensionality reduction), it is worth considering alternative methods as well.

- **Clustering and other analyses** One of the most common tasks of scRNAseq data analysis is to identify and classify cell populations (Andrews and Hemberg 2018). This task requires to assign cells to different groups (clusters), such that cells in the same clusters are similar and distinct from cells in other clusters. There is a great variety of clustering algorithms available, including k-means, hierarchical and consensus clustering (L. Peng et al. 2020). Benchmarking studies suggest that "no individual scRNA-seq clustering algorithm can capture true clusters and achieve optimal performance in all situations" (L. Peng et al. 2020).

Clustering is usually followed by cell annotation (i.e., assigning cell type to the identified clusters), which is done by finding cell type specific markers or using automatic (machine learning) tools such as CellTypist (Domínguez Conde et al. 2022). The subsequent steps in the analysis depend on the focus of the particular study and can include differential gene expression (dge) analysis, investigation of the dynamics of cellular systems (RNA velocity, pseudotime), inferring gene regulatory networks (GRNs), and more.

The described analysis pipeline enables insights into multicellular systems. It is evident that all downstream analyses are directly influenced by the initial steps of raw data processing, with mapping being particularly crucial in the context of this project. While mapping algorithms and tools differ, they all rely on the genome and its annotation (also referred as 'transcriptomic reference'), which directly impact their results. In the next section, I will further expand on this topic.

4.3 Genome annotation

The genome annotation process typically refers to the identification and mapping of genes within a given genome sequence (Guigó 2023). While the definition itself is straightforward, the process is highly complex. This is evident from the fact that, even more than 20 years after the first human genome assembly, human genome annotations are continuously updated with new transcripts and are expected to evolve further (Mudge et al. 2024). Figure 4.5 illustrates the changes in GENCODE annotation over time.

It is important to note that many medical and scientific research efforts rely on an accurate human gene list. Examples include genome-wide association studies (GWAS), which attempt to link genomic

variants to nearby genes; RNA-seq analysis; and exome sequencing projects that use capture kits targeting most known exons (Pertea et al. 2018).

Currently, the two most widely used genome annotations are GENCODE (Mudge et al. 2024) and RefSeq (O’Leary et al. 2015) (maintained by NCBI). Despite their status as mature genome references, they report different numbers of genes. For instance, RefSeq includes 20,078 protein-coding genes (NCBI 2025), whereas GENCODE contains 19,868 (Ensembl 2025). This discrepancy highlights the ongoing challenge of accurately annotating genomes.

Genome annotation process RNA sequencing (RNA-seq) is the primary tool used in genome annotation. Full-length RNA sequencing allows for the capture of RNA molecules, which, when aligned to a reference genome, help construct the genome annotation of a given species (Salzberg 2019). However, RNA-seq has limitations, the most significant being its inability to capture all RNA molecules. This poses particular challenges for detecting rare transcripts, which may either be treated as noise or not captured at all (Salzberg 2019). Therefore, bioinformatics tools are necessary to complement RNA-seq data (Guigó 2023), and most modern genome annotation pipelines integrate computational methods alongside sequencing data.

Computational genome annotation methods can be broadly categorized into two types: comparative annotation, which leverages the fact that protein-coding sequences tend to be more evolutionarily conserved, and *ab initio* annotation, which uses known sequence biases to predict genes (Guigó 2023). Despite significant advancements, genome annotation remains imperfect.

While the number of protein-coding genes is reaching a consensus — major databases report around 19,000 to 20,000 protein-coding genes — the number of non-coding genes is expected to increase in the future (Amaral et al. 2023). This is largely due to the structured nature of protein-coding genes (e.g., open reading frames, codon biases that skew nucleotide distributions), which makes them easier to detect using computational tools (Guigó 2023). Additionally, protein-coding genes have historically received greater attention because of their direct links to phenotype, leading to more experimental validation.

Current challenges in human genome annotation Despite extensive efforts, a fully comprehensive human genome annotation has not yet been achieved. Several factors contribute to this challenge.

Firstly, the complexity of the human genome itself presents significant obstacles. Compared to the total genome length, the number of genes is relatively low, and their sequences are interrupted by introns (Salzberg 2019). Additionally, bulk RNA sequencing often has relatively low sequencing depth, making it difficult to detect rare transcripts or those expressed in a cell-type-specific manner (Guigó 2023).

In principle, full-length single-cell RNA sequencing (scRNA-seq) could help overcome some of these limitations. However, it still has inherent biases introduced during library preparation, including those related to RNA processing status, post-transcriptional modifications, transcript length, cellular localization, and structural features (Guigó 2023). Short-read scRNA-seq typically provides higher throughput (Heumos et al. 2023), enabling the detection of rare transcripts. However, it does not capture full transcript structures, limiting its utility to supporting evidence, such as validating computationally predicted genes or indicating transcriptional activity at specific genomic locations.

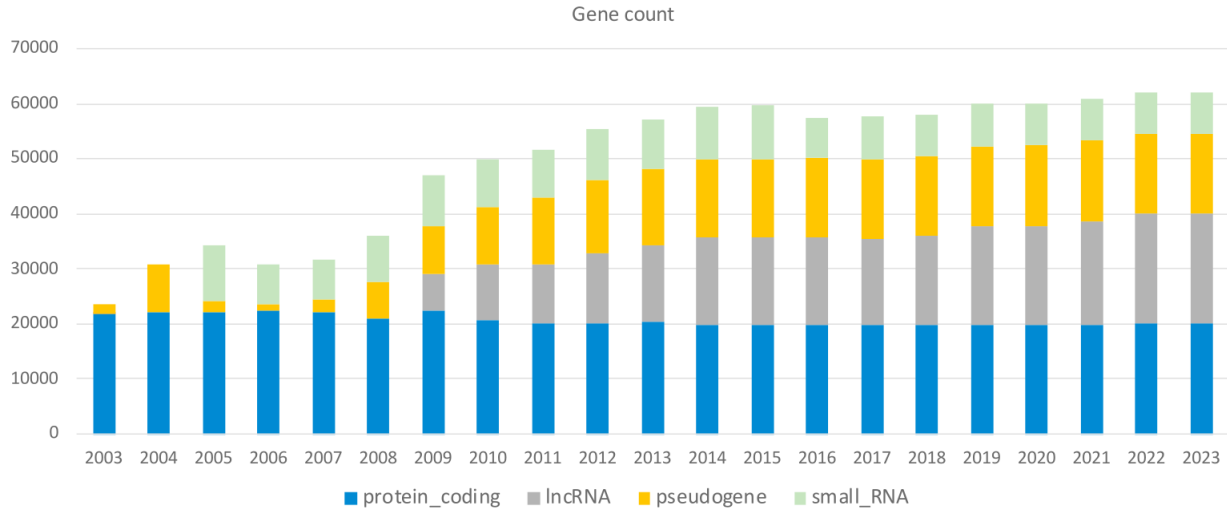


Figure 4.5: Number of different genes in the gencode annotation during time. Figure taken from paper by Guigó (2023)

Beyond technical limitations, there are ontological challenges, such as defining what constitutes a gene. While genes are often regarded as well-defined, discrete entities, the reality is more complex. Coding and non-coding transcripts frequently overlap in intricate arrangements with unclear boundaries, suggesting that transcripts may not be discrete, countable units but instead form a transcriptional continuum (Salzberg 2019). Additionally, the classification systems used in genome annotations may be partially artificial. For example, some pseudogenes, generally considered non-functional copies of functional genes, are transcribed and may have biological functions (Pei et al. 2012). Similarly, the distinction between protein-coding and non-coding genes is debated, as many protein-coding loci generate both coding and non-coding transcripts, and numerous long non-coding RNAs (lncRNAs) contain potentially coding open reading frames (ORFs) (Salzberg 2019).

Future perspectives of annotating genomes The size of eukaryotic genomes necessitates the use of automated methods for genome annotation. The accuracy of such methods depends directly on RNA capture technologies. If highly accurate and sensitive methods are developed, high-quality genome annotations can be expected (Salzberg 2019). However, at present, manual curation of human genome annotations remains essential due to the imperfections of both sequencing technologies and computational tools. Given the critical role of accurate gene annotation in both medical and scientific applications, continued improvements in annotation methods remain a priority.

4.4 Transcriptomic references for scRNAseq

Having a well-annotated genome does not solve all the challenges encountered during the mapping step. In simple terms, mapping algorithms determine the genomic location to which a sequence aligns, and if an annotated gene is present at that location, the read is assigned to that gene. However, there are cases where assigning a read to a gene is not straightforward, such as when a read maps to multiple locations or to a unique location where overlapping genes are annotated.

Multimappers There are two possible reasons for reads assigned to multiple genes: either they are mapped to multiple genome locations, or they are mapped to the genomic location containing overlapping features (genes). Such reads (commonly referred to as multimappers) are typically excluded from analysis (Almeida da Paz, Warger, and Taher 2024). However, as demonstrated by (Almeida da Paz, Warger, and Taher (2024)), this approach can introduce biases that affect downstream analyses. Currently, various methods exist to handle such cases (Deschamps-Francoeur, Simoneau, and Scott 2020), yet no entirely satisfactory solution has been established (Almeida da Paz, Warger, and Taher 2024).

The aforementioned methods primarily rely on computational strategies to resolve multimappers, but the issue can also be approached from a transcriptomic reference perspective. For example, in the case of CellRanger (the mapping software from 10x Genomics) the GENCODE reference is filtered based on gene types (e.g., retaining protein-coding genes while filtering out pseudogenes) (10x Genomics 2025). While this filtering improves the number of reads included in downstream analyses, there is still room for improvement.

Pool et al. (2023) proposed three key steps to enhance transcriptomic references:

1. Including reads mapped to intronic sequences in the analysis.
2. Extending the 3' ends of certain genes.
3. Resolving overlaps between specific genes.

The first suggestion is not new in scRNA-seq research. Concepts such as RNA velocity, which rely on the ratio of spliced to unspliced RNA (La Manno et al. 2018), demonstrate that including intronic reads can provide valuable information. Moreover, most mapping tools (e.g., STARsolo, CellRanger) offer options to align reads either to exonic regions only or to entire genes. The second suggestion is based on the observation that scRNA-seq data often exhibits peaks of reads just beyond the 3' ends of genes. While the exact biological reasons remain unclear — possibly due to imprecise annotations — it is reasonable to associate these reads with the nearest genes. The third suggestion addresses gene overlaps. Reads originating from overlapping regions are often unassigned to any gene, yet in some cases, they are more likely to come from one gene rather than another. Overlapping gene resolution seeks to correct this by modifying the transcriptomic reference, either by shortening or removing certain genes.

Although Pool et al. (2023) proposed a tool to implement these strategies, it has limitations. Some of its aspects remain debatable (e.g., threshold choices), others seem unnecessary (such as handling exon-intron distinctions when most alignment tools already provide this option), and the process still requires significant manual effort. Thus, there is still a need for a more comprehensive tool for enhancing transcriptomic references — a need that will be addressed in this thesis.

Reads mapped to the intergenic regions Some reads map to the genome but not to any annotated gene. While these could simply be noise in scRNA-seq data, there is also the possibility that they contain biologically relevant information. As will be shown in the results section, scRNA-seq datasets indeed contain such reads, and they are not entirely noise.

5. METHODS

5.1 Data Acquisition

scRNAseq datasets We analyzed datasets from four different human tissues: brain, blood, lung, and eye. All samples, except for two Peripheral Blood Mononuclear Cell (PBMC) datasets, were generated using the 10x Genomics v3.1 protocol. The two exceptions were prepared using the inDrops2 protocol. All protocols capture 3' ends of RNA molecules and uses short-read sequencing. The datasets are publicly available, with sources and additional details provided in Table 5.1.

sample name	tissue	cell count	donors	protocol	source
PBMC_10x	blood (PBMC)	5000	1	10x v3.1	10x genomics
PBMC_10x_2	blood (PBMC)	10000	1	10x v3.1	10x genomics
PBMC_10x_3	blood (PBMC)	10000	1	10x v3.1	10x genomics
PBMC_indrops	blood (PBMC)	2000	1	indrops2	Juzenas et al. 2025
PBMC_indrops_2	blood (PBMC)	9000	1	indrops2	Juzenas et al. 2025
brain	brain	6000	1	10x v3.1	Siletti et al. 2023
brain_2	brain	7000	1	10x v3.1	Siletti et al. 2023
eye	retina	10000	6	10x v3.1	Menon et al. 2019
eye_2	peripheral retina	2500	1	10x v3.1	Voigt et al. 2019
eye_3	peripheral retina	2500	1	10x v3.1	Voigt et al. 2019
lung_2	lung	5000	1	10x v3.1	Mould et al. 2021
lung_5	lung	5000	1	10x v3.1	Mould et al. 2021
lung_7	lung	4500	1	10x v3.1	Mould et al. 2021
lung_8	lung	5000	1	10x v3.1	Mould et al. 2021

Table 5.1: Datasets summary.

Genome and Transcriptomic references The human genome assembly GRCh38 was used in this project, downloaded from the [Ensembl website](#).

Four transcriptomic references were analyzed (see Table 5.2 for details). To ensure compatibility with the genome FASTA file, chromosome name prefixes ('chr') were removed from references that included them, as the genome file does not use these prefixes. This modification was performed using basic Linux command-line tools. Additionally, for the NCBI reference, chromosome names were converted to Ensembl-style notation using a custom Python script (e.g., 1, 2, 3 instead of NC_000001.11, NC_000002.12, NC_000003.12, etc.).

For references that do not have 'gene' entries, such entries were added (entry that spans all the components of a particular gene).

reference name	source	version
10x	10x website	2024-A
GENCODE	Gencode website	47
RefSec (NCBI)	NCBI website	GCF_000001405.40
lnc	LNCipedia website	5.2

Table 5.2: References used in this project.

Gene Prediction Tracks and Conservation Scores Gene predictions (tracks 'AUGUSTUS', 'Geneid genes', 'Gescan genes', 'SGP genes', 'SIB genes') and genome conservation scores ('phast-Cons100way' track) were downloaded from [UCSC genome browser](#).

ATAC data The open chromatin regions (bed format) in PBMCs was downloaded from [10x website](#).

5.2 Computational Tools and Environment

All analyses were performed on a high-performance computing (HPC) cluster running a Linux environment. Software packages and tools were managed using Conda. The exact Conda environment specifications (YAML file) can be found on [GitHub repository](#). Besides tools managed by conda, also basic command line tools were used (e.g. *awk*, *wc*, *grep*, *sed*, *uniq*, *sort* and similar). The specific functions and parameters used are described in the following sections, where the general analysis pipeline will be explained.

5.3 Data processing pipeline

The pipeline was implemented using GNU MAKE and the full scripts are available on the [GitHub](#), here only general description of the workflow and used tools is provided. The general scheme of the pipeline (used both for extracting intergenic regions and enhancing transcriptomic reference) can be seen in Figure 5.1, while the description is given below:

1. Map reads with initial transcriptomic reference.
2. Take unassigned (and uniquely mapped) reads.
3. Split into intersecting and intergenic reads.
 - (a) For intersecting:
 - i. Resolve overlapping genes that have unassigned reads (if possible).
 - ii. From the second reference and further: add genes to the original GTF that contain unassigned reads and do not overlap with entries from the original one.
 - (b) For intergenic:
 - i. Cluster.
 - ii. Filter-out relatively small clusters (custom threshold).
 - iii. For the first reference only: filter-out AT-rich reads.

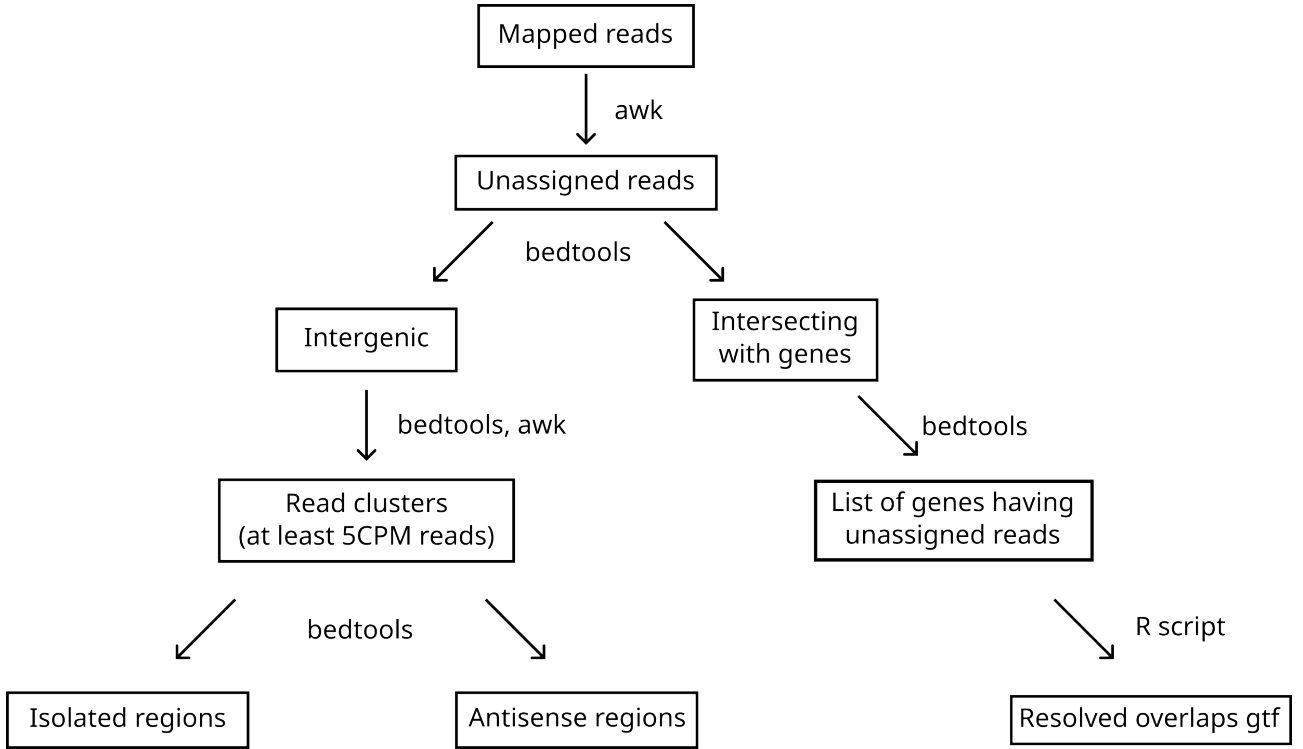


Figure 5.1: Scheme of the pipeline. This scheme provides only general logic and main tools of data processing. The pipeline itself is more complex, e.g., using multiple transcriptomic references required looping over some of those steps.

- iv. For reads that are left, repeat from the beginning with the next reference.
 - v. For the last reference only: clusters that start just after 3' ends are assigned to genes (i.e., extend genes).
4. Create final GTF and map initial sequences to it.
 5. Create list of large (>5CPM) intergenic clusters.

Mapping and filtering bam files Reads were mapped using STARsolo (Kaminow, Yunusov, and Dobin 2021), both in the case of mapping from fastq and bam input files. Parameters were used the same for all samples (see [GitHub repository](#)), except the ones regarding barcode structures (differences include number of barcodes, lengths of barcode/UMI sequences, and their positions in the primer). Filtering unassigned reads was done using *awk*, taking those that have valid barcode (i.e. filtering out those reads that have barcode length not equal to the defined length) and were not assigned to any gene (i.e. had 'GN:Z:-' tag). Also, only uniquely mapped reads were taken (tag 'NH:i:1').

Classifying reads as 'intergenic' or 'intersecting' and reads clustering Intersections of unassigned reads with references were checked using *bedtools intersect* command. Those that intersect with genes were classified as 'intersecting', and those that do not – as 'intergenic'. Intersections were checked in strand-specific manner (*bedtools* '-s' flag), i.e. regions on different strands were treated as not intersecting, even if they share overlapping genomic coordinates (see Figure 5.2). The intergenic reads were clustered using *bedtools merge*, again in a strand-specific manner.

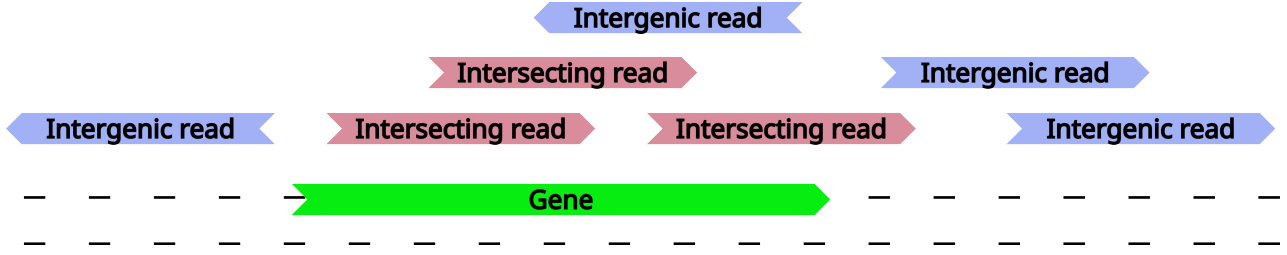


Figure 5.2: Unassigned read classification as intergenic or intersecting. Reads are defined as intersecting if they overlap with known genes and are on the same strand. Remaining reads are defined as intergenic.



Figure 5.3: Classification of intergenic regions as antisense or isolated.

Manipulating transcriptomic references The overlapping gene resolving and construction of enhanced transcriptomic reference was done using custom R script, particularly *rtracklayer* library.

5.4 Intergenic regions

Extracting intergenic regions The intergenic reads were extracted as described in the previous section (i.e. taken unassigned reads that do not overlap with any reference used) and merged into clusters using *bedtools merge* command. Clusters were filtered based on cluster size, requiring at least 5 reads per million (CPM), where CPM represents the normalized read count per million of total primary reads in the dataset (i.e. $1CPM = 1 \times sample_size / 1000000$). The filtering was done using *awk*.

Cluster locations were adjusted using *deeptools* (to compute coverage) and custom python script, to avoid reads containing long introns making the intergenic clusters very wide. To accomplish this, for each intergenic region the maximum coverage location was found and extended to include neighbouring regions that had at least $max_value/2$ coverage. In such way, intergenic regions were adjusted to cover only 'peaks' of reads.

The lists acquired from all samples were then merged using *bedtools merge* function. This combined list then was filtered to contain only regions detected at least in all samples of one group ('eye', 'lung', 'brain', 'PBMC_10x' or 'PBMC_indrops'). Such filtering still allows to include tissue-specific regions in the combined list, while removing those that are not persistent across the same type of samples. Additionally, for each entry it was determined whether it overlaps with predicted genes from UCSC gene prediction archive (using *bedtools merge*) and distances to the closest genes were found (*bedtools closest*). Intergenic regions that were located on the opposite strand of known genes, were classified as *antisense*, while those that were not were classified as *isolated* (see Figure 5.3 for illustration).

This filtered combined list was then converted into GTF format and unassigned reads from each sample were mapped using this combined intergenic annotation.

Analysis of cell-gene matrices The matrices produced by STAR were filtered, allowing cells that have sufficient number of reads (thresholds selected manually), and only genes that were expressed

in at least 3 cells. Also cells were filtered based on mitochondrial gene count (allowing up to 10% of mitochondrial gene counts in a cell). Doublets were filtered using *scrublet*. Afterwards, matrices were normalized (using *normalize_total* function from *scanpy* package) and log-transformed (i.e. for each entry $x = \log(x + 1)$, *log1p* function from *scanpy*).

Then only highly-variable genes (min_mean=0.0125, max_mean=3, min_disp=0.5) were selected. For the clustering, principal components (PCs) were computed and optimal number of them were selected based on elbow rule (manually). For blood samples, cells were annotated automatically using CellTypist (for other samples, annotation was skipped, as it was not in the main focus of this project). Then visualizations were made using UMAP embeddings (functions from the same *scanpy* package).

All the scripts with descriptions can be found in the [GitHub repository](#).

Correlations Spearman’s rank correlation between intergenic region and gene on the opposite strand was checked using custom python script. Formula for Spearman’s correlation is given by:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the ranks of corresponding values and n is the number of observations.

5.5 Enhancing transcriptomic reference

Three aspects were considered while enhancing transcriptomic reference:

- **Gene overlaps** – some of genes are overlapping, resulting in unassigned reads from those overlapping regions.
- **3’ end extensions** – for some genes, there are clusters of reads just after their 3’ ends, and Pool et al. (2023) suggested to include those reads by extending 3’ ends for some genes.
- **Different annotations** – while most of the annotations agree on most genes, there are small fraction of genes that are present in some references but not in others. Hence, for the enhancing transcriptomic reference, several transcriptomic references were used.

Overlaps The criteria for the resolving of overlapping genes are following:

1. **Gene type:** prefer protein coding genes over other types, lncRNA over remaining (e.g. pseudogenes).
2. **Level:** some annotations have ‘level’ field, indicating if the gene is verified (score 1), manually annotated (score 2) or automatically annotated (score 3). Lower ‘level’ score was preferred.
3. **Intersection types:** if 5’ end gene is overlapping with 3’ end of gene, 5’ end of gene was shortened.

Particularly, genes types was divided into three groups: protein coding, long non-coding RNAs and other, and each class was assigned score (see Table 5.3). The overall priority score was defined as

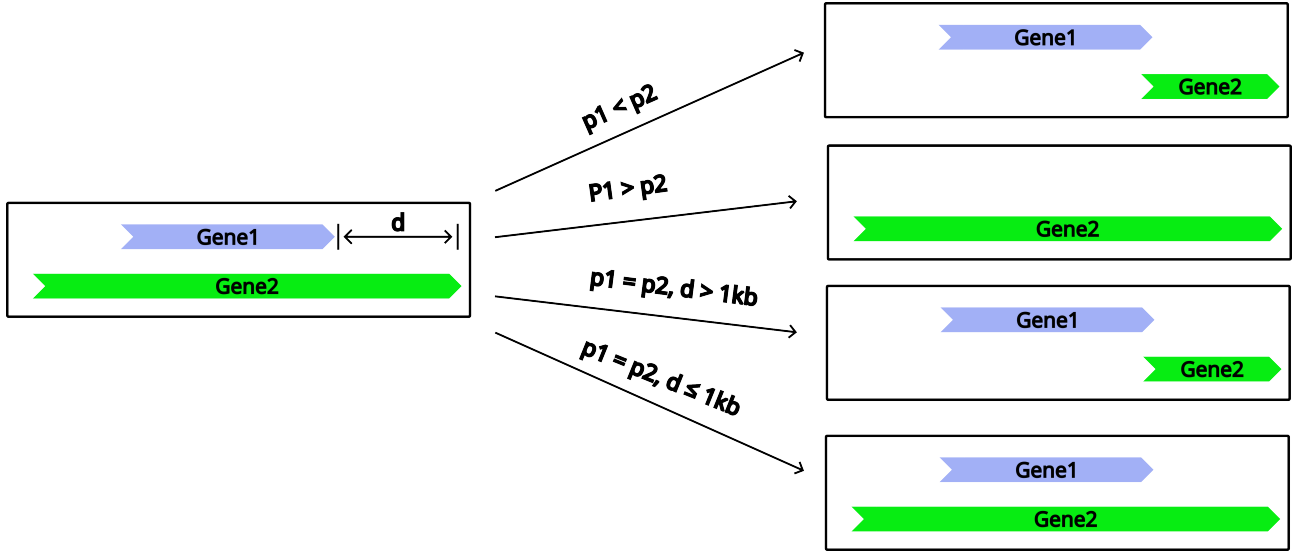


Figure 5.4: Strategy for resolving overlaps. $p1$ and $p2$ are priority scores of *gene1* and *gene2* (lower score means higher priority), while d is distance between 3' ends of those genes.

sum of this gene type score and 'level' field. The lower this score is, the higher priority gene has. The intersection type (whether 3' end is overlapping or not) was taken into account only in cases when priority scores were equal, and distances between 3' ends of overlapping genes were greater than 1kb (see Figure 5.4 for graphical explanation). If possible, lower priority genes were shortened, otherwise they were deleted.

Class	Score	Gene Types
protein coding	0	protein_coding, IG_C_gene, IG_D_gene, etc.
long non-coding RNA	10	lncRNA, lincRNA
other	20	snRNA, snoRNA, pseudogene, etc.

Table 5.3: Priority scores assigned based on gene type.

3' end extensions Unassigned reads clusters (of size at least 1CPM) that were close to the 3' of genes (closer than 100bp) were assumed to originate from those genes. Hence annotations for those genes were expanded to cover those clusters.

Multiple annotations The initial transcriptomic reference (taken from 10x Genomics website) was enriched with entries from other annotations. Particularly entries from GENCODE, RefSeq and Incpedia transcriptomic references were added to the initial reference, if they overlapped with unassigned reads and did not overlapped with genes already present in the reference. This enrichment was done sequentially, i.e. firstly the reference was updated with entries from GENCODE annotation, then resulting reference was updated with entries from RefSeq annotation and finally from Incpedia annotation.

6. RESULTS

6.1 Intergenic regions

Fourteen datasets from various tissues (blood, brain, eye, and lung) were analyzed. These datasets contained between 11% and 29% unassigned uniquely mapped reads (as shown in Table 6.1) when using the 10x reference. Additionally, 6% to 18% of total reads were intergenic, meaning they did not overlap with any reference used. Given the size of the datasets, such a proportion of reads could potentially contain biologically relevant information.

Sample	Total Reads	Unassigned	Intergenic	Demultiplexed	Unassigned	Intergenic
PBMC_10x	182330834	15.95%	11.37%	70305137	18.51%	12.86%
PBMC_10x_2	496387931	18.52%	14.26%	195948926	21.29%	16.32%
PBMC_10x_3	368640939	18.59%	14.34%	167804193	20.99%	16.17%
PBMC_indrops	112932507	11.12%	6.5%	8176578	22.86%	10.25%
PBMC_indrops_2	471705924	14.87%	8.36%	98063027	25.52%	13.73%
brain	206360627	16.87%	10.67%	143916177	19.02%	12.07%
brain_2	122556503	22.32%	15.98%	95031864	23.77%	17.00%
eye	375397270	28.02%	18.97%	161882445	31.87%	21.60%
eye_2	140981808	29.35%	14.85%	69146995	31.43%	15.99%
eye_3	161261977	29.46%	18.86%	68157696	32.11%	20.58%
lung_2	511080104	15.99%	12.07%	269118747	17.98%	13.33%
lung_5	452105505	14.49%	10.74%	256353198	16.19%	11.83%
lung_7	524095146	18.41%	14.34%	236157103	20.64%	15.79%
lung_8	342092138	14.53%	10.66%	217801055	16.07%	11.67%

Table 6.1: Statistics of unassigned and intergenic reads per sample. Unassigned reads were counted after mapping with 10x reference. Intergenic reads here are those that do not intersect with any reference used. 'Demultiplexed' column shows number of total reads after demultiplexing using UMIs, and following 'unassigned' and 'intergenic' percentages were computed compared to this number.

Analyzing intergenic regions For each sample, intergenic regions containing a sufficient number of unassigned reads were identified, as described in the methods section (with a threshold set at 5 CPM). The term "intergenic" is used here in a strand-specific manner, meaning an "intergenic" region may be located on the opposite strand of a known gene.

To determine whether these intergenic regions contain biologically meaningful information, cells in each sample were clustered based solely on reads from these regions. As shown in selected examples in Figure 6.1, clustering was observed and roughly corresponded to clustering based on the standard ("10x") annotation. This indicates that biologically meaningful information is indeed present in these

intergenic regions. Consequently, further analyses were performed on these regions.

The lists of intergenic regions from all samples were combined and filtered based on the number of samples in which they were detected, resulting in a final set of 2,590 intergenic regions. Of these, 147 did not overlap known genes on the opposite strand and will be referred to as "isolated". Those located on the opposite strand of known genes will be referred to as "antisense".

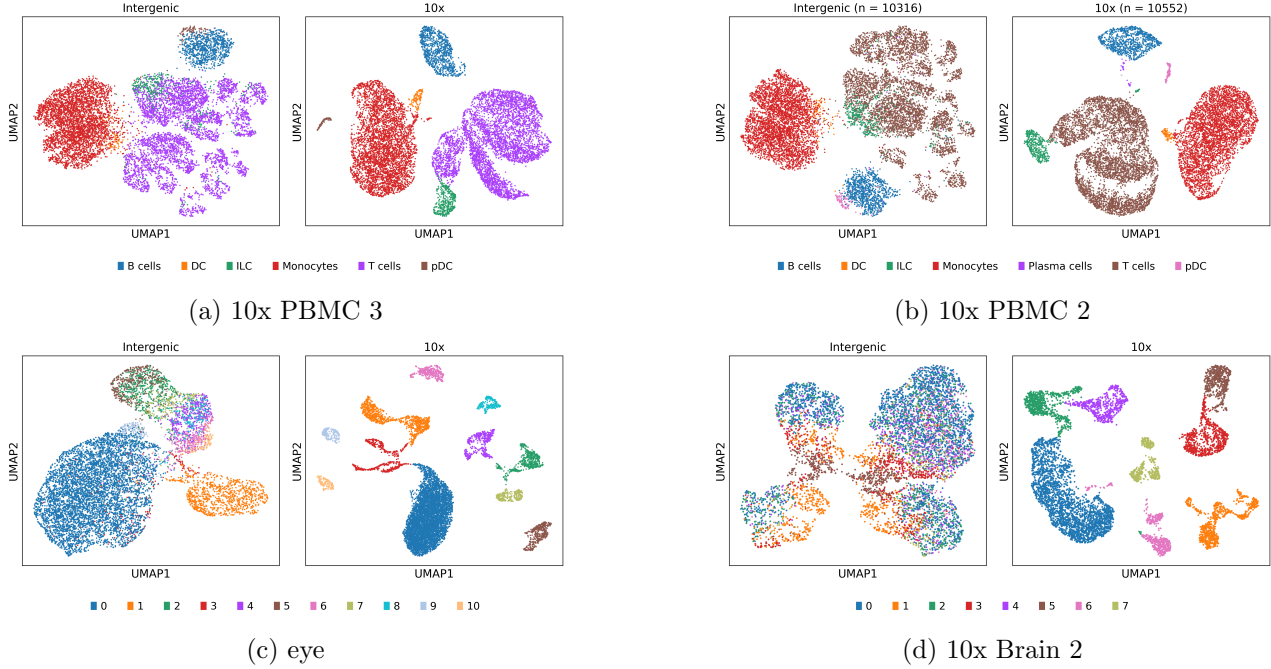


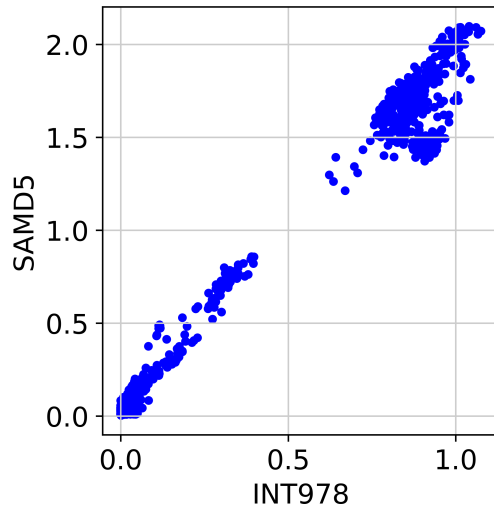
Figure 6.1: Comparison of clustering using standard annotation ('10x' reference) and using only defined intergenic regions. As can be seen, for some samples intergenic regions are sufficient for rough clustering, for others (e.g. 'brain.2' sample), noise gives some clustering artefacts. Nevertheless, this rough clustering implies that these unassigned read contain biological information.

A-rich sequences near the intergenic regions 3' end If there is poly-A region near the 3' end of the intergenic region, then the presence of those peaks can be explained by mis-priming, i.e. that would suggest that our intergenic region is not necessarily the actual 3' end of the novel transcript, but may be in the middle of it.

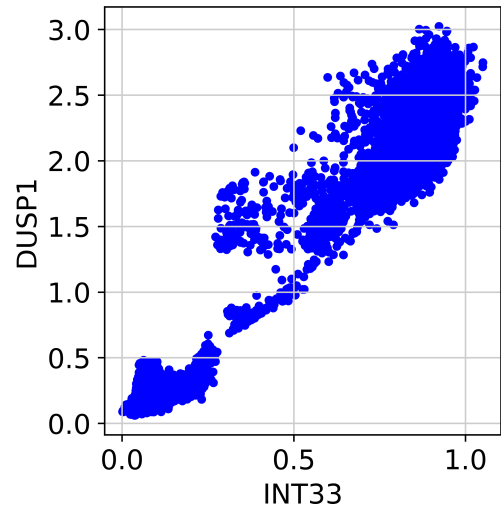
This is indeed the case for the most of reported intergenic regions, see Figure 6.2 for visual example and some statistics.

Correlations Another aspect to assess is whether expression of those intergenic regions correlate with the expression of nearby genes. If so, this may be due to several factors:

- If the region correlates with gene on the same strand:
 - Technical issues – the genome annotation might be not comprehensive and not include some longer transcripts (that would span intergenic region as well as the gene with which correlation is observed).
 - Biological reasons – genes in close proximity tend to be coexpressed due to the shared chromatin structure.



(a) Correlation can be clearly seen in INT33 and DUSP1 normalized expressions (Spearman correlation 0.68) in the PBMC_10x.2 sample, both of them are on the same strand.



(b) Correlation of INT978 and SAMD5 normalized expressions (Spearman correlation 0.85) in the eye_3 sample, intergenic region is on the opposite strand of the gene.

Figure 6.3: Examples showing correlations between intergenic regions and nearby genes.

2. Open chromatin: Are open chromatin regions present upstream? Such regions often indicate active transcription.
3. Conservation score: Coding regions tend to be more evolutionarily conserved than non-coding regions.
4. Gene predictions: Do these regions overlap with predicted genes identified by computational tools?

Differential expression analysis To assess differential expression, all data samples were first clustered. For *PBMC_10x* samples, clustering and cell annotation were performed using CellTypist. For other samples, the Leiden algorithm was used, with manually chosen parameters to ensure a comparable number of clusters across similar datasets. For instance, all *lung* samples were clustered into five groups, aligning approximately with visible UMAP structures. The UMAP visualizations, colored by clusters, can be found in Appendix 14.1.

After filtering those regions that were differentially expressed (thresholds were used 0.01 for adjusted p-value and 0.5 for 'logfoldchange'), the 29 differentially expressed isolated intergenic regions were found. Some examples can be seen in Figure 6.4.

Open chromatin Open chromatin regions may provide further evidence of transcriptional activity when located upstream. However, distances to the nearest known genes must also be considered, as upstream open chromatin sites may not be associated with the intergenic regions but rather with other genes.

For each intergenic region, distances to the nearest open chromatin sites and upstream genes (strand-unspecific) were calculated. Of the 147 isolated intergenic regions, 66 had open chromatin regions closer than any known upstream genes. To ensure that these sites were not merely associated

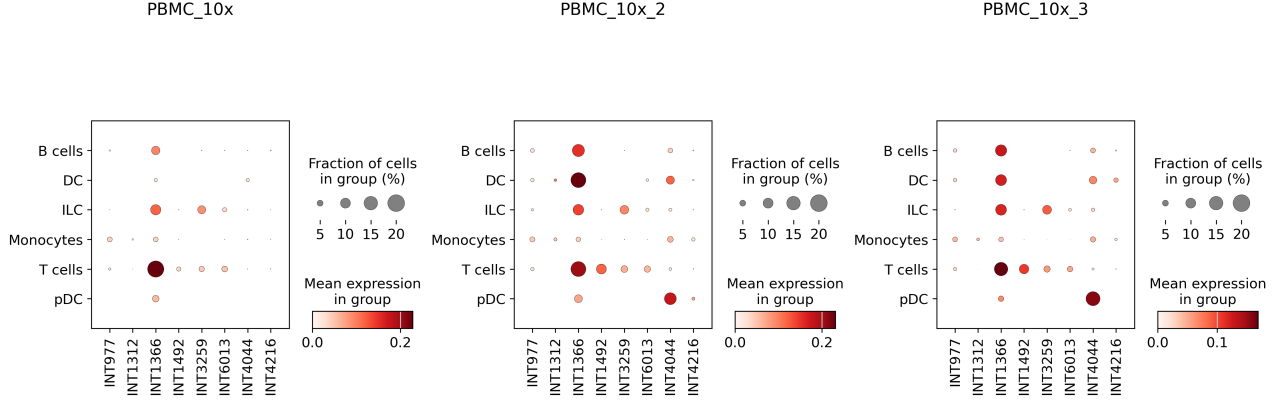


Figure 6.4: Examples of differentially expressed isolated intergenic regions in *PBMC_10x* samples.

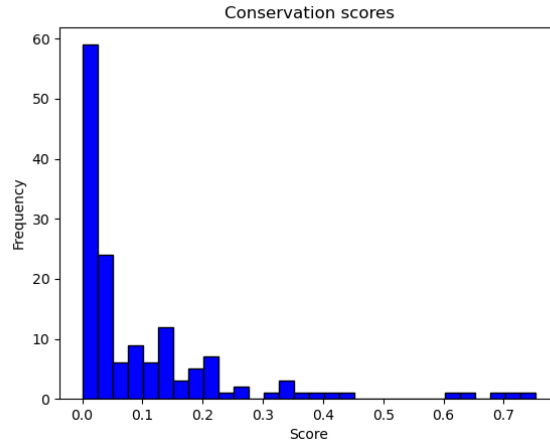


Figure 6.5: Histogram showing conservation scores in the isolated regions.

with distant genes, an additional filter was applied, requiring that the closest open chromatin region be at least 2,000 bp from the nearest gene. This yielded 40 intergenic regions.

It should be noted, that PBMC ATAC data was used for open chromatin regions, thus this feature should be taken with precaution for regions derived from other datasets.

Conservation scores The conservation score shows evolutionary conservation and is computed comparing genomic sequences from different species. A high conservation score in an intergenic region suggests potential functionality, providing additional evidence for biological relevance. To identify such regions, those with conservation scores above 0.6 was defined as conserved (there were 5 of such isolated regions). The overall conservation scores were quite low, suggesting that if the isolated intergenic genes originate from unannotated genes, they are probably human specific. The histogram showing conservation scores in isolated regions is given in the Figure 6.5.

Gene predictions To assess whether isolated intergenic regions overlap with predicted genes, UCSC gene prediction archives were examined, focusing on overlaps with 3' ends of predicted genes.

Of the identified regions, 12 overlapped with predicted genes. However, three of these did not overlap with the 3' ends of predicted genes, which would be expected given that the data was generated

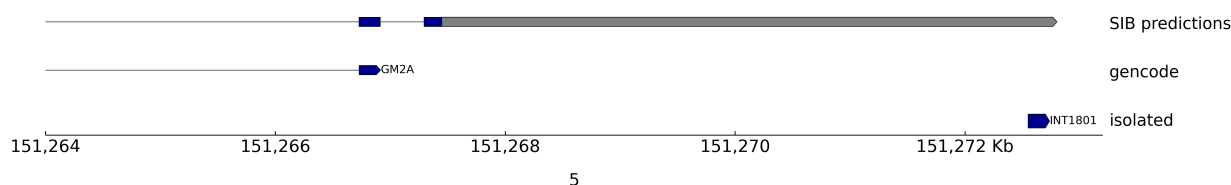


Figure 6.6: Example of gene which is predicted to have some longer transcripts than are present in the gencode and ncbi annotations. The intergenic region is located after the 3' end of gene GM2A, however, SIB prediction tool suggest that longer transcripts should be included in the annotation of this gene.

using 3' end sequencing methods. Five predicted genes were extensions of known genes, while the remaining four corresponded to genes absent from RefSeq and NCBI references but supported by our data.

Combining supporting features In total, four features were examined as supportive evidence for the biological significance of these intergenic regions: conservation, open chromatin presence, differential expression, and gene predictions. Based on these, four sublists of isolated intergenic regions were generated:

- Conserved regions
- Regions associated with open chromatin
- Differentially expressed regions
- Regions overlapping predicted genes

Notably, the list of conserved regions had only one element (INT1216) in common with the other lists (particularly, prediction list). The predicted gene list had two regions in common with the differential expression list (INT196, INT1525) and one in common with the open chromatin list (INT4147). All three of these predicted genes were extended versions of already annotated genes (see example in Figure 6.6). The differential expression and open chromatin lists had 11 regions in common (INT1312, INT1426, INT1492, INT1554, INT1609, INT1993, INT2209, INT349, INT368, INT4216, INT6013). No regions appeared in more than two lists.

6.1.2 Antisense intergenic regions

Interpretation of antisense intergenic regions is more complicated. The features such as open chromatin, AT-richness or conservativeness are not strand specific, meaning that if they would be present, they could not be attributed with certainty for our intergenic regions. The differential expression can be checked, however, even if the antisense intergenic region is differentially expressed, it still can be an artifact, originating from differentially expressed transcripts. Hence from the features checked for the isolated intergenic regions, only real supporting evidence would be computational gene prediction.

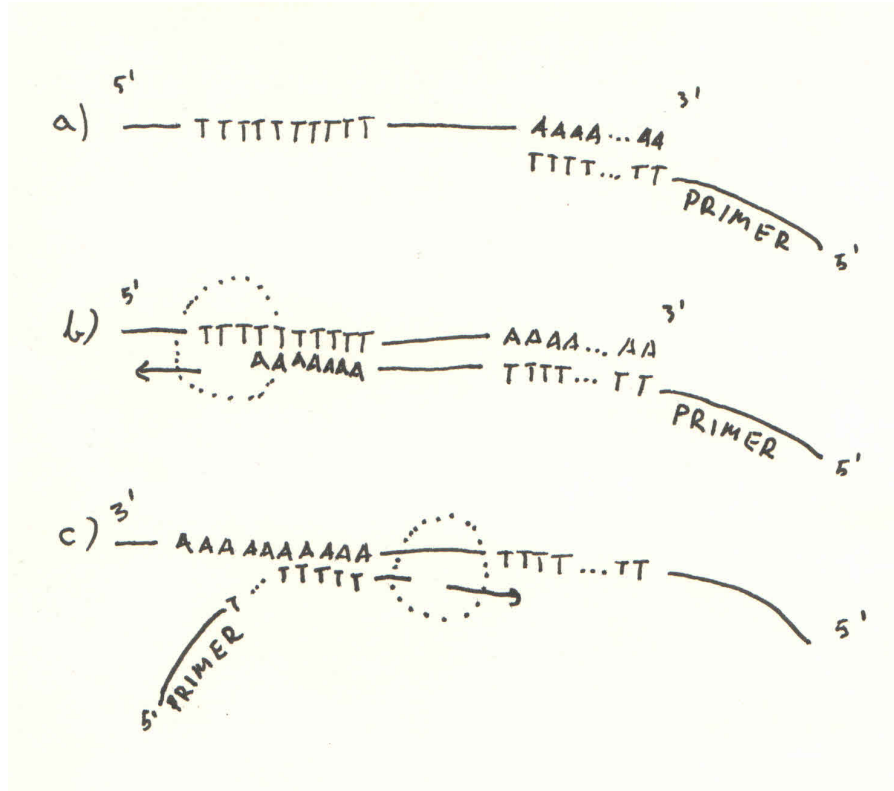


Figure 6.7: Scheme of poly-A mispriming resulting in reads mapped to the opposite strand compared to the original gene. a) RNA is captured by a poly-A tail with a primer containing a poly-T sequence. b) Reverse transcriptase attaches to the primer and synthesizes cDNA from the RNA template. c) If the RNA-cDNA complex becomes unstable and dissociates, another primer may bind to the poly-A stretches on the cDNA, if present. This would result in reversed cDNA, causing the reads to be mapped to the antisense strand.

Gene predictions As in the case of isolated intergenic regions, antisense intergenic regions were checked for overlaps with predicted genes from UCSC prediction archive. There were 49 such regions found, out of them 35 were in the 3' ends of predicted genes. 9 of those 35 predicted genes were longer version of already annotated genes in RefSeq or GENCODE references. The all list can be found in the Appendix [14.1](#).

Poly-A segments Having in mind that majority of the antisense intergenic regions have poly-A stretch after 3' end, and some of them strongly correlates with genes on the opposite strands, it is possible that they are artefacts from the library preparation. Particularly, it could happen that poly-T primer binds to poly-A stretch of cDNA produced by other primer, and in such way inversed transcripts are produced, which are later mapped to antisense strands of known genes (see Figure [6.7](#)).

6.2 Enhancing transcriptomic reference

Keeping in mind that unassigned reads are also present in gene locations, we attempted to resolve overlapping genes in the annotations so that reads from these regions could be included in the downstream analysis. For overlap resolution, a simple set of rules was applied, primarily favoring protein-coding genes over others and considering the level field in the transcriptomic annotations (more details can

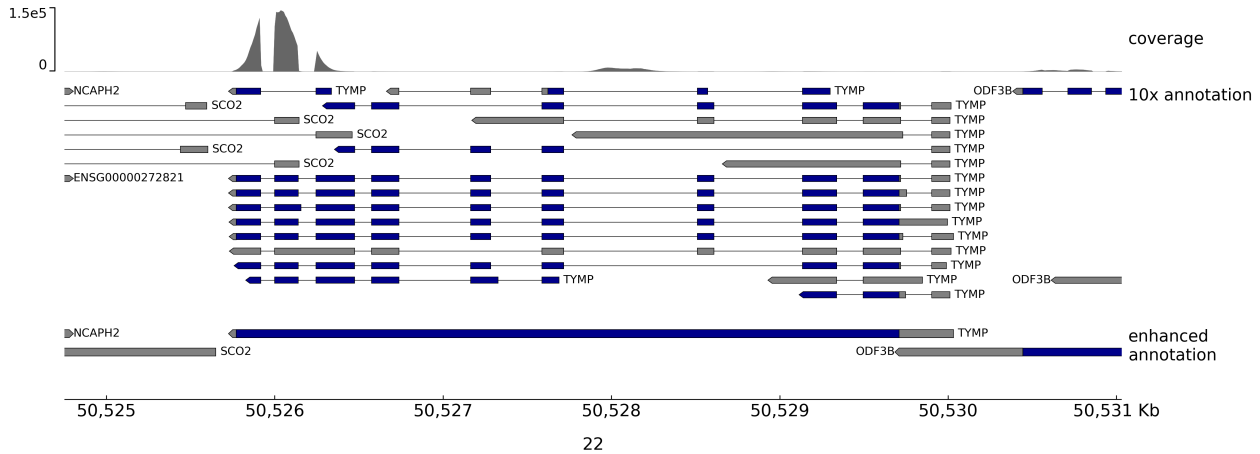


Figure 6.8: Example of successful resolution of overlapping genes TYMP and SCO2. Both genes are protein coding, however TYMP is preferred since it has lower *level* value. Based on the exon structures, it can clearly be seen that those reads in the overlapping area indeed originate from TYMP gene. The 10x annotation track is expanded to highlight exon structures.

be found in Methods 5.5).

In addition to resolving overlaps, the initial transcriptomic reference (taken from 10x Genomics) was enriched with entries from other transcriptomic references (RefSeq, GENCODE, lncRNAPedia), ensuring that new entries did not overlap those already present. Finally, gene entries with reads near their 3' ends were extended.

This enhanced reference resulted in an average increase of 1.2% more reads in the cell-gene matrices. While this increase is relatively small, it enabled the detection of approximately 350 additional genes that were undetected using the standard transcriptomic reference (an example is given in Figure 6.8).

Although the overall data volume increased, it is not straightforward to conclude that data quality improved. Overlap-handling rules can be debated, particularly regarding their applicability in all cases.

For example, genes are expressed in certain cell types or states regardless of their level classification (whether verified, manually annotated, or automatically annotated). While, in general, verified or manually annotated genes are more frequently expressed than automatically annotated ones (which are likely less explored and detected at lower rates overall), this pattern does not hold in all situations. The same consideration applies to gene classification by type.

Regarding the gene extensions, in most cases they increase the gene expression by relatively small fraction of counts (see Figure 6.9 for example). Even though some additional information is gained, when compared with the total amount of data, this gain is very small (e.g., in *PBMC_10x_2* sample, it constituted only 1.76% of gained reads, or 0.017% of total reads).

Overall, while statistical differences are evident, their impact on biological interpretations remains uncertain and likely depends on the specific downstream analysis. For instance, cell clustering is generally unaffected, and a visual inspection of UMAPs does not reveal significant differences between datasets generated using the enhanced and standard references. However, in certain cases, even a small number of additional genes could significantly influence the analysis, meaning that this enhanced reference could help uncover relevant biological processes.

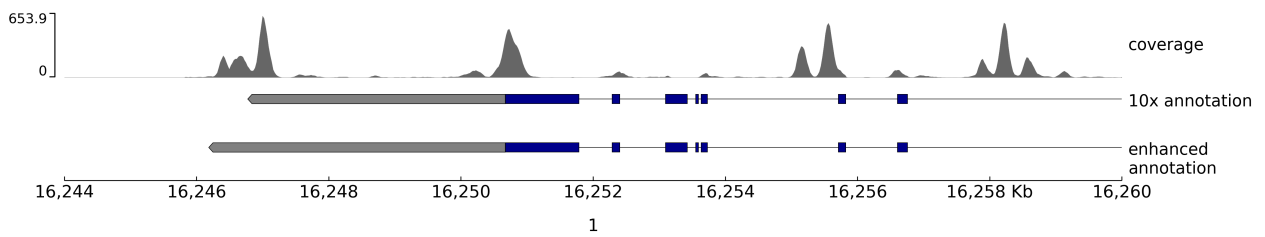


Figure 6.9: Example of gene extension (*PBMC_10x_2* dataset). In this case gene FBX042 is extended to include reads close to its 3' end. However, the amount of reads gained is tiny compared when compared with total count of this gene (the number of counts of this gene increased approximately by 3% in the *PBMC_10x_2* dataset).

7. DISCUSSION

In the results chapter was shown that the unassigned reads constitute significant part of total scRNAseq data. Such reads could roughly be categorized into two classes: overlapping with genes in the genome annotation and not.

Those that are not overlapping with genes, were clustered and classified either as antisense (located on the opposite strand of the known genes) or isolated (remaining ones). Antisense regions constituted absolute majority of defined intergenic regions. The observations about correlations and poly-A stretches allowed us to make hypothesis that those antisense reads are artifacts generated during reverse transcription reaction. Isolated reads, on the other hand, can originate from unannotated genes, especially those, that were supported by some additional evidence, such as differential expression or ATAC-seq data.

For those overlapping with known genes, the main reason for the being unassigned is overlaps between genes (i.e. if a read maps to the region where two genes are present, mapping algorithm does not know, for which to assign). The suggested pipeline to resolve these overlaps (mainly based on gene type) allowed to have up to 100% more genes in the downstream analysis. It should be noted, however, that while overall number of reads and number of total captured genes ending up in the cell-gene matrices increases, there are genes with decreased total counts after resolving overlaps. This highlights the fact that human genome is very complex, and there is no simple method to resolve overlaps when mapping short reads. There are other tools (!!!!!!give examples!!!!!!). None of them are agreed to be gold standards, though.

It should be noted, that datasets used were generated using **short-read 3' end** protocols. Therefore, datasets prepared using different methods might have different artifacts present. However, the new gene candidates remain valid, and they should be verified by some additional experiments.

As for the reverse transcription reaction artefacts, it is worth considering performing this reaction in lower temperatures, as RNA-cDNA complex is more stable in such temperatures.

8. CONCLUSIONS

- The single cell RNA sequencing data contains reads that are mapped to the genome, but not assigned to any gene. The amount of such reads varied between 11-29% in datasets analyzed.
- The unassigned reads mapped to the antisense of known genes strands constitutes major part of unassigned reads are likely artifacts produces during reverse transcription reaction. It should be noted that this applies only to short-read 3' end sequencing protocols.
- The unassigned reads mapped to locations without known genes present on any strand might be originating from unannotated genes. This thesis suggested a list of 147 such candidates. The type of the experiments (short-read 3' end sequencing) has not allowed to infer global structures of those pottentionally novel genes.
- Some of the unassigned reads that were mapped to the locations containing overlapping genes might be included into downstream analysis by resolving gene overlaps in the transcriptomic annotation. The proposed approach allowed to have approximately 1.2% more reads in the cell-gene matrices used in the downstream analysis.

9. RECOMMENDATION

The unassigned reads are present in the scRNA-seq datasets, and some of them contain biologically meaningful information. Hence it is recommended not to discard those unassigned reads completely. Particularly, those that are located in isolated regions (not on the opposite strand of known genes) might be unannotated genes. This should especially be important in the case when new or less explored organisms are studied, as their transcriptomic annotation might be not comprehensive.

Also, resolving overlaps in the genome annotation might help to have more genes in the downstream analysis. Therefore, if one wants to have as much information as possible about the sample of the study, one should consider resolving overlaps in the transcriptomic references.

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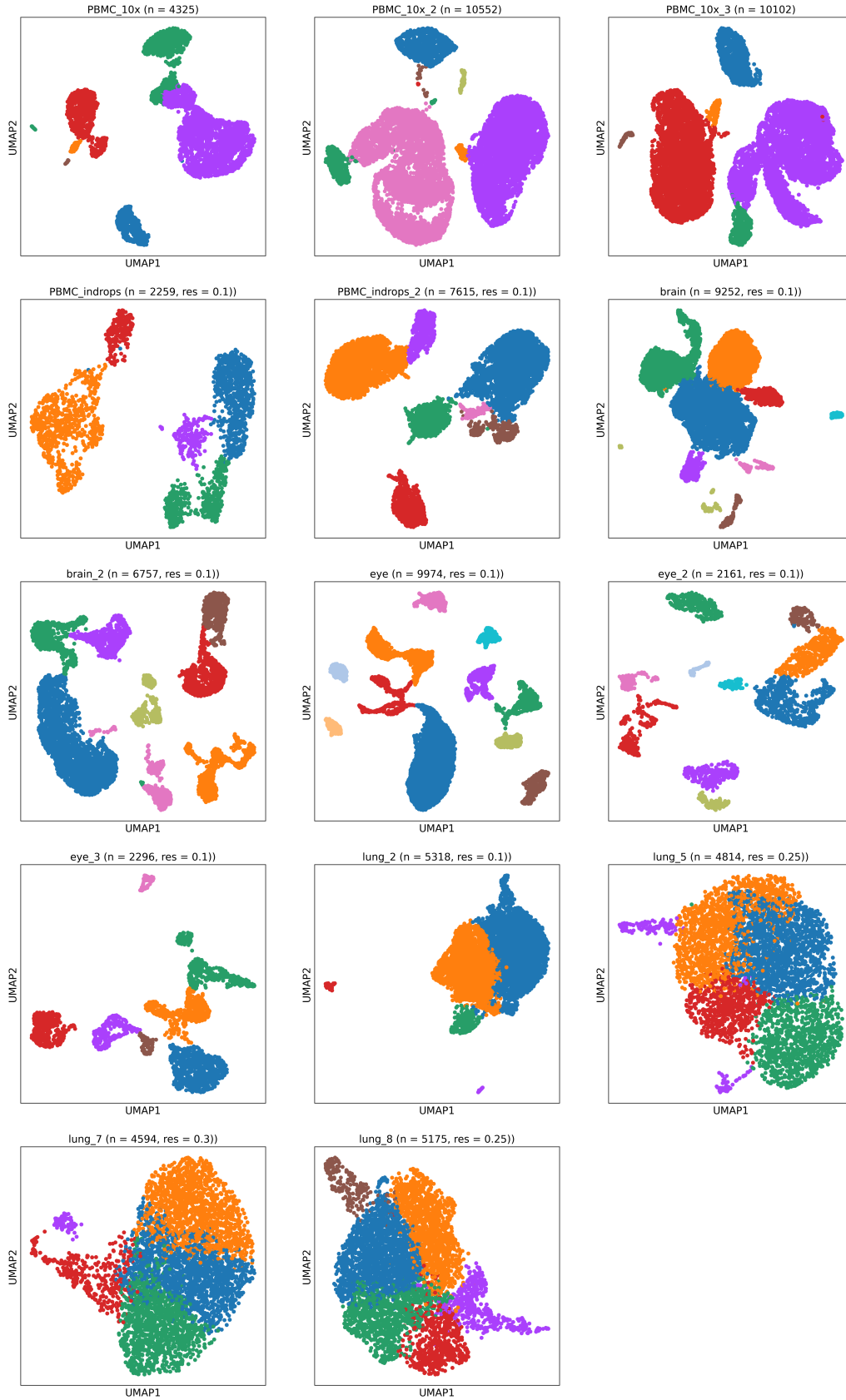
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12. SUMMARY

13. SUMMARY IN LITHUANIAN

14. APPENDICES

Figure 14.1: Clusterings of datasets. 'n' stands for number of cells in the plot, 'res' for the resolution parameter of leiden algorithm. *PBMC_10x* samples are colored by CellTypist annotations.



Name	Genomic coordinates	Prediction tool	Overlaps with 3' region of predicted gene
INT7	6:24718850-24719100	SIB	Yes
INT33	5:172767250-172767500	SIB	Yes (DUSP1)
INT134	15:41280550-41280800	SIB	Yes (extends a bit after)
INT218	5:61409150-61409400	SPG	No (5' end)
INT285	10:110898200-110898550	SIB	Yes (BBIP1)
INT327	1:169132100-169132350	SIB	Yes (NME7)
INT347	14:75277700-75278000	Gscan	No
INT382	12:11892250-11892500	Gscan	Yes
INT386	6:26215200-26215450	SIB	Yes (H2BCB?)
INT438	4:39713300-39713600	SIB	Yes (almost all (short) gene spanned)
INT533	8:130110900-130111150	SIB	Yes
INT617	15:85733950-85734200	SIB	Yes
INT621	9:75148850-75149400	SIB	Yes (OSTF1)
INT651	1:28579100-28579350	SIB	Yes (TRNAU1AP)
INT654	6:34382850-34383050	SIB	No (but short gene)
INT828	8:133035800-133036050	SIB	Yes (SLA)
INT859	2:71373500-71373750	SIB	No (slight overlap on 5' end)
INT898	17:51190400-51190700	SIB	Yes (extends after)
INT903	8:22613400-22613650	SIB	Yes
INT949	9:111693050-111693400	SIB	Yes
INT972	15:64954800-64955350	SIB	Yes
INT984	18:63291900-63292200	Geneid	Yes
INT1028	5:50414200-50414500	SIB	Yes
INT1155	1:36296050-36296300	SIB	No (5' end)
INT1231	10:19890700-19891050	Gscan	Yes
INT1402	5:142786700-142786950	SIB	No (5' end)
INT1440	2:235746450-235746700	SIB	Yes
INT1445	9:40885650-40885900	SIB	Yes
INT1548	7:50342700-50342950	Gscan	No (exon in the middle)
INT1616	17:30895450-30895700	SIB	No (5' end)
INT1749	3:172333550-172333750	SIB	No (5' end)
INT1931	17:82522100-82522350	SIB	Yes
INT2065	10:103606150-103606400	SIB	Yes
INT2158	14:49636100-49636350	SIB	No (5' end, DNAAF2)
INT2194	7:130521250-130521500	SIB	No (but gene is short)
INT2371	1:14632650-14632900	Geneid	No (middle exon)
INT2380	22:33853900-33854150	Geneid	Yes
INT2825	13:45300050-45300300	Geneid	No (middle exon)
INT2979	4:98929350-98929600	SIB	No (5' end, EIF4E)
INT3328	2:184604250-184604500	SIB	Yes
INT3429	17:48172750-48173150	Augustus	Yes
INT3504	15:25332950-25333150	SIB	Yes (UBE3A)
INT3849	3:121432200-121432400	SIB	Yes
INT3868	17:31448500-31448700	SIB	Yes
INT4458	5:44820700-44820900	SIB	Yes (MRP530)
INT4743	20:3188900-3189050	SIB	Yes (DDRKG1)
INT5002	11:92232750-92232950	SIB	Yes
INT5315	20:3183750-3184050	SIB	Yes
INT5329	13:95603000-95603200	SIB	Yes (extends after)

Table 14.1: Antisense intergenic regions overlapping with predicted genes from UCSC gene prediction archive. The gene in the brackets shows if the predicted gene is extended version of already annotated genes.